

Reference PDF Content

Save the following text as a PDF named HDS5_Analysis_Reference.pdf and place it in a new folder named assets in the same directory as your Python script.

1. Analysis Workflow

- 1. User Input Validation:** The application accepts total design discharge (Q_{total}) and distributes it equally across the specified number of barrels to determine Q_{barrel} .
- 2. Geometry Check:** The app enforces dimensional constraints based on shape (Box vs. Circular) and calculates Area (A), Wetted Perimeter (P), Hydraulic Radius (R), and Flow Velocity (V).
- 3. Inlet Control Calculation:** Evaluates both unsubmerged and submerged states using FHWA HDS-5 nomograph polynomial equations. The governing inlet headwater (HW_i) is the maximum of these two values.
- 4. Outlet Control Calculation:** Determines the energy grade line at the outlet (h_o) based on Tailwater (TW) and Critical Depth (d_c), then applies full-flow friction and entrance losses to determine HW_o .
- 5. Governing State Selection:** Compares HW_i and HW_o (unless overridden by the user) to establish the controlling headwater elevation.

2. Governing Equations (FHWA HDS-5 & TxDOT HDM Ch. 8)

- Discharge Intensity:

$$DI = \frac{Q_{barrel}}{AD^{0.5}}$$

- Inlet Control (Unsubmerged Form 1):

$$\frac{HW_i}{D} = \frac{H_c}{D} + K [DI]^M - 0.5S$$

- Inlet Control (Submerged):

$$\frac{HW_i}{D} = c [DI]^2 + Y - 0.5S$$

- Outlet Control Energy Balance:

$$HW_o = \left(1 + k_e + \frac{29n^2L}{R^{1.33}} \right) \frac{V^2}{2g} + h_o - LS$$

(Where $h_o = \max(TW, \frac{d_c + D}{2})$)

3. Polynomial Coefficients & Entrance Losses (Table 28 & 29)

Headwall Type	Shape	K	M
TxDOT 0° Wingwalls	Box	0.061	0.75
TxDOT 45° Wingwalls	Box	0.061	0.75
Circular Conc. (Square)	Circ.	0.0098	2.0
Circular CMP (SET)	Circ.	0.0210	1.33